CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -1

AIM: - Study of various Network commands used in Linux and Windows:

Aim

To study and execute various network commands available in Linux and Windows operating systems to obtain information about network configuration, test connectivity between computers, identify network routes, resolve domain names, and troubleshoot common network problems.

Procedure

1. Start the computer and open the command-line interface. In Windows, open **Command Prompt** by typing cmd in the Start menu. In Linux, open the Terminal application. These command-line tools allow users to execute network commands and examine the output.
2. Execute the ipconfig command in Windows to display the network configuration of the computer. It provides information such as the IPv4 address, subnet mask, and default gateway for the available network adapters.

ipconfig

In Linux, use the following command to display the IP address and network interface information:

ip addr

Observe the IP address assigned to the computer and identify the active network interface.

3. Execute the ipconfig /all command in Windows to obtain detailed network configuration information. This includes the physical or MAC address, DHCP status, DNS servers, and other adapter details. In Linux, the ip link command can be used to examine network interface information, including MAC addresses.

4. Use the ping command to test connectivity with another computer or a destination host. For example, execute the following command:

ping 127.0.0.1

The address 127.0.0.1 is the loopback address used to test the local TCP/IP network stack. Observe the replies and response times displayed by the command.

5. Execute the ping command with the IP address of another computer on the same network. For example:

ping 192.168.1.2

Replace the example address with the actual IP address of the destination computer. If the destination is reachable and permits ICMP Echo Requests, the command displays reply messages. Otherwise, it may display a timeout or an error.

6. Execute the hostname command to display the name of the current computer.

hostname

The hostname helps identify a computer on the network and can be useful when checking network connectivity between multiple systems.

7. Execute the getmac command in Windows to display the MAC addresses of network adapters.

getmac

The MAC address is a hardware address associated with a network interface. In Linux, the ip link command can be used to identify the MAC address of each network interface.

8. Execute the tracert command in Windows to examine the route taken by packets toward a destination.

tracert google.com

In Linux, use the traceroute command if it is installed. The command displays intermediate network hops between the source computer and the destination. Some hops may not respond because of filtering or network configuration.

9. Use the nslookup command to obtain DNS information for a domain name.

nslookup google.com

Observe the DNS server used for the query and the IP address information returned for the domain. This command helps troubleshoot problems related to domain name resolution.

10. Execute the netstat command in Windows to display active network connections and listening ports.

netstat

To display additional connection and port information, use:

netstat -an

Observe the local and remote addresses, port numbers, and connection states shown in the output. These details help examine network activity on the computer.

11. Execute the arp -a command to display the Address Resolution Protocol (ARP) table.

arp -a

The ARP table contains mappings between IPv4 addresses and MAC addresses that the computer has learned on its local network. The entries help explain how a device identifies the local hardware address associated with an IPv4 address.

12. Execute the route print command in Windows to display the IP routing table.

route print

In Linux, use the following command:

ip route

Examine the routing entries and identify the default route. The routing table helps the operating system determine where packets should be forwarded to reach local and remote networks.

13. In Linux, execute the ping command with a known destination and use ip route to inspect the route configuration. Additional commands such as ss -tuln can be used to display listening TCP and UDP ports. These commands provide information similar to some of the network inspection functions available in Windows.
14. Record the output of the commands and compare the information obtained from Windows and Linux. Note that command availability and output formatting may differ between operating systems. Some commands may require administrative privileges or additional software.
15. If a command fails, verify its syntax, the destination address, and the network connection. Check the IP configuration, default gateway, DNS settings, and firewall restrictions where appropriate. A failed ping does not always mean that a destination is completely unreachable, since ICMP traffic may be blocked.

Windows commands

C:\Users\Balag>hostname

BalaGanesh

C:\Users\Balag>ipconfig /all

Windows IP Configuration

Host Name . . . . . : BalaGanesh

Primary Dns Suffix . . . . . :

Node Type . . . . . : Mixed

IP Routing Enabled. . . . . : No

WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :

Description . . . . . : VirtualBox Host-Only Ethernet Adapter

Physical Address. . . . . : 0A-00-27-00-00-0E

DHCP Enabled. . . . . : No

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::aa4a:775:1de1:1757%14(Preferred)

IPv4 Address. . . . . : 192.168.56.1(Preferred)

Subnet Mask . . . . . : 255.255.255.0

Default Gateway . . . . . :

DHCPv6 IAID . . . . . : 118095911

DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F

NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. . . . . : F8-FE-5E-3A-70-56

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

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Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. . . . . : FA-FE-5E-3A-70-55

DHCP Enabled. . . . . : No

Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz

Physical Address. . . . . : F8-FE-5E-3A-70-55

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::b810:29c9:55e9:3a65%17(Preferred)

IPv4 Address. . . . . : 10.83.96.83(Preferred)

Subnet Mask . . . . . : 255.255.255.0

Lease Obtained. . . . . : 26 August 2026 14:38:14

Lease Expires . . . . . : 26 August 2026 15:38:11

Default Gateway . . . . . : 10.83.96.20

DHCP Server . . . . . : 10.83.96.20

DHCPv6 IAID . . . . . : 167312990

DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F

DNS Servers . . . . . : 10.83.96.20

NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

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   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : F8-FE-5E-3A-70-59
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek Gaming GbE Family Controller
   Physical Address. . . . . . . . . : 2C-58-B9-DE-5F-5F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Users\Balag>arp -a

Interface: 192.168.56.1 --- 0xe
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
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```
Interface: 10.83.96.83 --- 0x11
  Internet Address      Physical Address      Type
  10.83.96.20           4e-ec-85-ec-ff-15     dynamic
  10.83.96.255          ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\Balag>netstat -r
===========================================================================
Interface List
 14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter
 11...f8 fe 5e 3a 70 56 ......Microsoft Wi-Fi Direct Virtual Adapter
  4...fa fe 5e 3a 70 55 ......Microsoft Wi-Fi Direct Virtual Adapter #2
 17...f8 fe 5e 3a 70 55 ......Intel(R) Wi-Fi 6E AX211 160MHz
 19...f8 fe 5e 3a 70 59 ......Bluetooth Device (Personal Area Network)
  2...2c 58 b9 de 5f 5f ......Realtek Gaming GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      10.83.96.20      10.83.96.83     35
       10.83.96.0    255.255.255.0         On-link       10.83.96.83    291
```

```
     10.83.96.83  255.255.255.255        On-link      10.83.96.83    291
    10.83.96.255  255.255.255.255        On-link      10.83.96.83    291
       127.0.0.0        255.0.0.0        On-link        127.0.0.1    331
       127.0.0.1  255.255.255.255        On-link        127.0.0.1    331
  127.255.255.255  255.255.255.255        On-link        127.0.0.1    331
    192.168.56.0    255.255.255.0        On-link      192.168.56.1    281
    192.168.56.1  255.255.255.255        On-link      192.168.56.1    281
  192.168.56.255  255.255.255.255        On-link      192.168.56.1    281
       224.0.0.0        240.0.0.0        On-link        127.0.0.1    331
       224.0.0.0        240.0.0.0        On-link      192.168.56.1    281
       224.0.0.0        240.0.0.0        On-link       10.83.96.83    291
  255.255.255.255  255.255.255.255        On-link        127.0.0.1    331
  255.255.255.255  255.255.255.255        On-link      192.168.56.1    281
  255.255.255.255  255.255.255.255        On-link       10.83.96.83    291
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
 14    281 fe80::/64                On-link
 17    291 fe80::/64                On-link
 14    281 fe80::aa4a:775:1de1:1757/128
                                    On-link
 17    291 fe80::b810:29c9:55e9:3a65/128
```

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On-link

1 331 ff00::/8 On-link

14 281 ff00::/8 On-link

17 291 ff00::/8 On-link

===

Persistent Routes:

None

C:\Users\Balag>nslookup www.google.com

Server: UnKnown

Address: 10.83.96.20

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:482d:7700::

2001:4860:4826:7700::

2001:4860:4828:7700::

2001:4860:482b:7700::

2001:4860:4827:7700::

2001:4860:4829:7700::

2001:4860:482a:7700::

2001:4860:482c:7700::

142.251.152.119

142.251.156.119

142.251.154.119

142.251.151.119

142.251.153.119

142.251.157.119

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142.251.150.119

142.251.155.119

C:\Users\Balag>ping google.com

Pinging google.com [142.251.43.46] with 32 bytes of data:

Reply from 142.251.43.46: bytes=32 time=14ms TTL=114

Reply from 142.251.43.46: bytes=32 time=233ms TTL=114

Reply from 142.251.43.46: bytes=32 time=59ms TTL=114

Reply from 142.251.43.46: bytes=32 time=328ms TTL=114

Ping statistics for 142.251.43.46:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 14ms, Maximum = 328ms, Average = 158ms

C:\Users\Balag>route print

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Interface List

14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter

11...f8 fe 5e 3a 70 56 ......Microsoft Wi-Fi Direct Virtual Adapter

4...fa fe 5e 3a 70 55 ......Microsoft Wi-Fi Direct Virtual Adapter #2

17...f8 fe 5e 3a 70 55 ......Intel(R) Wi-Fi 6E AX211 160MHz

19...f8 fe 5e 3a 70 59 ......Bluetooth Device (Personal Area Network)

2...2c 58 b9 de 5f 5f ......Realtek Gaming GbE Family Controller

1...........................Software Loopback Interface 1

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IPv4 Route Table

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Active Routes:

Network Destination Netmask Gateway Interface Metric

0.0.0.0 0.0.0.0 10.83.96.20 10.83.96.83 35

10.83.96.0 255.255.255.0 On-link 10.83.96.83 291

10.83.96.83 255.255.255.255 On-link 10.83.96.83 291

10.83.96.255 255.255.255.255 On-link 10.83.96.83 291

127.0.0.0 255.0.0.0 On-link 127.0.0.1 331

127.0.0.1 255.255.255.255 On-link 127.0.0.1 331

127.255.255.255 255.255.255.255 On-link 127.0.0.1 331

192.168.56.0 255.255.255.0 On-link 192.168.56.1 281

192.168.56.1 255.255.255.255 On-link 192.168.56.1 281

192.168.56.255 255.255.255.255 On-link 192.168.56.1 281

224.0.0.0 240.0.0.0 On-link 127.0.0.1 331

224.0.0.0 240.0.0.0 On-link 192.168.56.1 281

224.0.0.0 240.0.0.0 On-link 10.83.96.83 291

255.255.255.255 255.255.255.255 On-link 127.0.0.1 331

255.255.255.255 255.255.255.255 On-link 192.168.56.1 281

255.255.255.255 255.255.255.255 On-link 10.83.96.83 291

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Persistent Routes:

None

IPv6 Route Table

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Active Routes:

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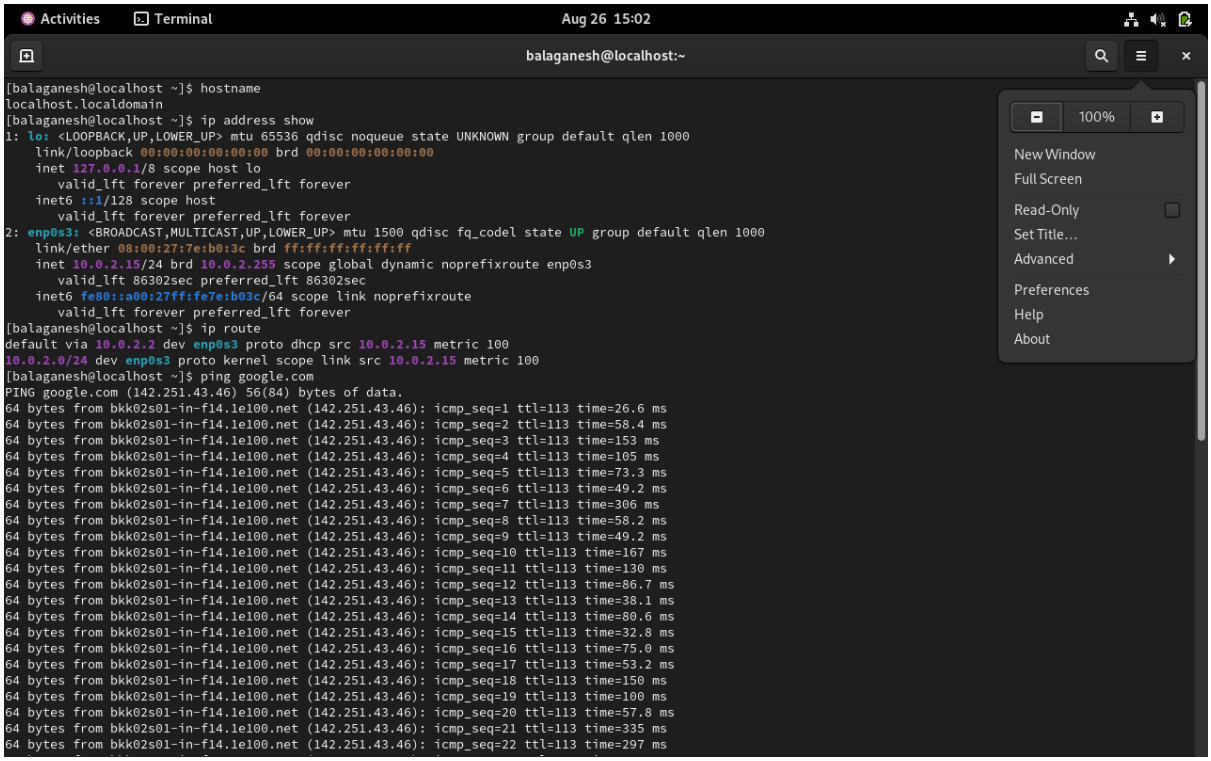
```
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
 14    281 fe80::/64                On-link
 17    291 fe80::/64                On-link
 14    281 fe80::aa4a:775:1de1:1757/128
                                    On-link
 17    291 fe80::b810:29c9:55e9:3a65/128
                                    On-link
  1    331 ff00::/8                 On-link
 14    281 ff00::/8                 On-link
 17    291 ff00::/8                 On-link
===========================================================================
```

Persistent Routes:

None

C:\Users\Balag>tracert google.com

Linux commands



Student observation answers

1. **Command used to check reachability:** ping
2. **Command used to find hops:** tracert in Windows / traceroute in Linux
3. **Command to display IP configuration:** ipconfig in Windows / ip address in Linux
4. **Command to display TCP port status:** netstat
5. **Command to modify IP configuration in Linux:** ip address / ip route

Result

Thus, various network commands used in Windows and Linux were studied and executed. The commands were used to examine IP configuration, test connectivity, identify host and MAC addresses, inspect routing information, resolve domain names, display active connections, and examine ARP entries. The experiment demonstrated how command-line network utilities help administrators understand network configuration and troubleshoot common connectivity problems.